

Statement of intent

Baha Khammari · PhD position, auction-based support schemes for renewable energy
LIDAM Energy Group, CORE, UCLouvain

Why this doctorate

Energy economics has followed me through both degrees at the University of Cologne, where I finish an M.Sc. this month with a thesis in applied microeconometrics. What kept me in it is that decarbonising our economies will be a monumental effort, reaching past capital into how our societies work, and that almost all of it runs through instruments governments design rather than technology alone. Which technologies get built, and at what cost to the public, is settled by those rules, and that is what I want to spend four years on.

Why CORE

My training sits on the econometrics side of that question, and it has taken me to the edge of what econometrics alone answers. Reading an auction rule properly needs graduate industrial organization and game theory, which I do not have yet and have started on anyway, working through Jehle and Reny in my own time. It is why I am writing to CORE rather than a department that does one of these things, because a centre built around operations research, economics and econometrics is where both halves of the question already sit. The other half of what I want is people. Everything I have written so far I wrote alone, and an Erasmus semester in Rotterdam showed me I work better in a room where nobody shares my background, explaining what I am working on and hearing what others are. That is most of what a research group does, and I want it among your doctoral students and visiting researchers.

What I would bring

My thesis, graded 1.0 and worth 30 of 120 ECTS, asked whether a national scholarship programme in Brazil raised formal-sector wages, on a panel I built from administrative records, five million linked employer-employee observations across 558 microregions, 2005 to 2019, in R through BigQuery. Callaway and Sant’Anna (2021) group-time effects with doubly-robust inference across exposure bins put the gain at 3.04 per cent in low-exposure regions, distinguishable from zero at the 10 per cent level, on a non-monotonic dose-response whose sign two-way fixed effects reverses. What that taught me is that a policy can look inert until you allow for how differently places were exposed to it, which is the same problem a support scheme poses. Since the 2016 state aid framework member states have arrived at technology-neutral, technology-specific and mixed designs at different dates, so the variation is staggered adoption of a design rather than of a policy, and the AURES II database records every renewable auction in the EU and UK from 2011 to 2021. I would use it to estimate how joint bidding shifts the awarded portfolio and the rent per megawatt hour against member states not yet treated. Behind the econometrics sit advanced microeconomics on strategic thinking and competition with Axel Ockenfels at 1.3, 1.0 in mathematics, and a model I calibrated in MATLAB.

Thank you for considering my application. I could start in October 2026 and would be glad to talk the project through whenever it suits you.

Referees

Prof. Dr. Pia Pinger, University of Cologne, chair of my master’s thesis.
pia.pinger@uni-koeln.de

Jun.-Prof. Dr. Oliver Ruhnau, University of Cologne and EWI, examiner in *Energy Markets and Regulation*.
ruhnau@wiso.uni-koeln.de

Baha Khammari · b.e.khammari@gmail.com · +49 1520 9016956